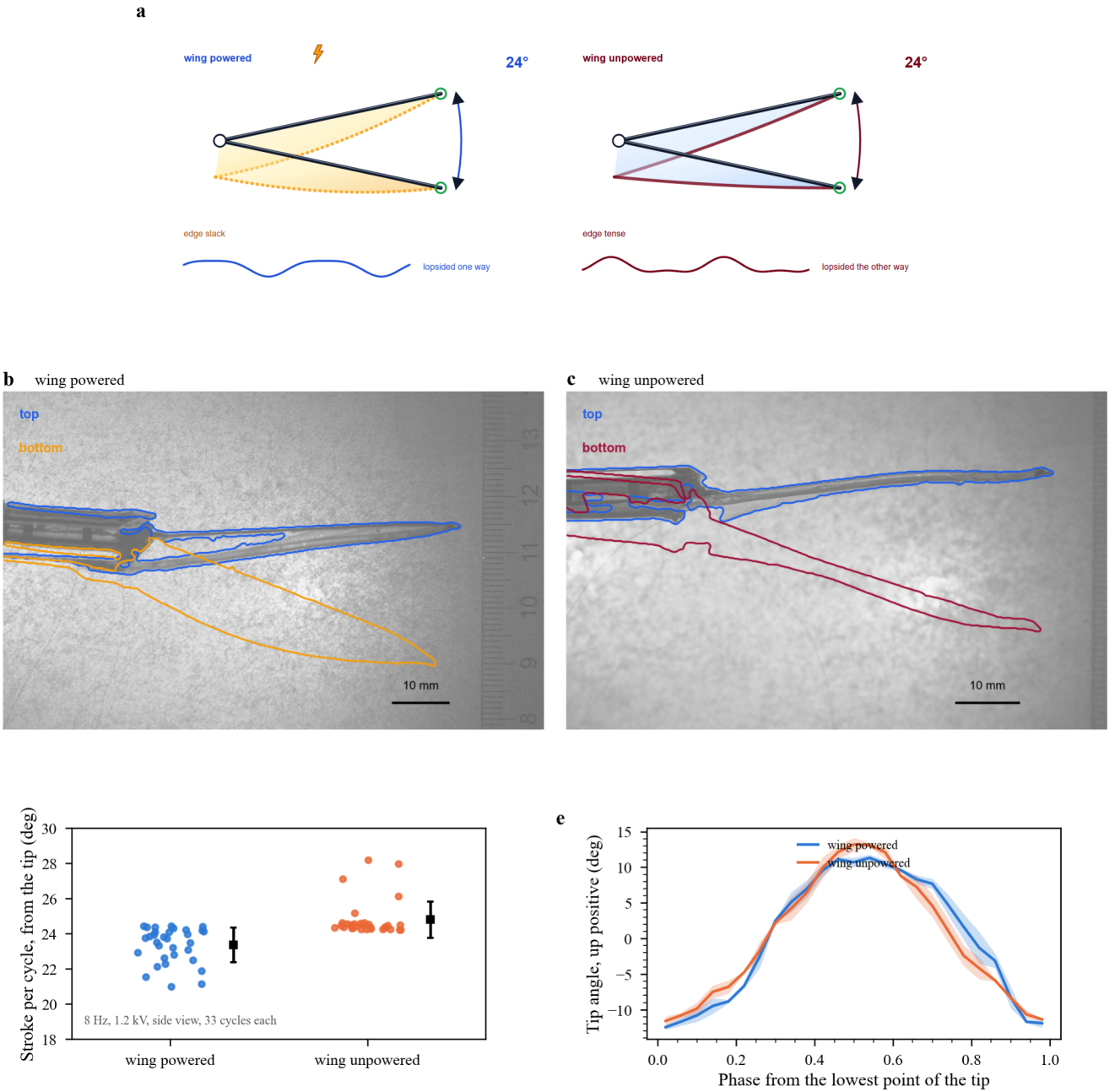

a
wing powered
24°
wing unpowered
24°
edge slack
lopsided one way
edge tense
lopsided the other way
b
wing powered
top
bottom
10 mm
c
wing unpowered
top
bottom
10 mm
d
Stroke per cycle, from the tip (deg)
30
28
26
24
22
20
18
8 Hz, 1.2 kV, side view, 33 cycles each
wing powered
wing unpowered
e
Tip angle, up positive (deg)
15
10
5
0
−5
−10
wing powered
wing unpowered
0.0
0.2
0.4
0.6
0.8
1.0
Phase from the lowest point of the tip